

JEE ADVANCED - 6 | PAPER - 2

JEE 2022

Date	Maximum Marks	Duration
16th January, 2022	186	3 Hours

PAPER SCHEME

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
- Section 1** contains 2 types of questions [**Type A & Type B**].

Type A contains **8 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Type B has **2 List-Match Sets**. Each List Match set has to **Two (02) Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple-Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Section 2** contains **6 Numerical Value Type Questions**. The answer to each of the question is a numerical value. For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.** (Example: 6, 81, 1.50, 3.25)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION - 1 | Type - A

- This section consists of **Eight (08)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks:** +4 If only (all) the correct option(s) is(are) chosen
 - Partial Marks:** +3 If all the four options are correct but **ONLY** three options are chosen
 - Partial Marks:** +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct
 - Partial Marks:** +1 If two or more options are correct but **ONLY** one option is chosen, and it is a correct option
 - Zero Mark:** 0 If none of the options is chosen (i.e. the question is unanswered)
 - Negative Marks:** -1 In all other cases.

SECTION - 1 | Type – B

- **Type B** has **2 List-Match Sets**. Each List Match set has to Two (02) Multiple Choice Questions. Each List-Match question has two lists: List-I and List-II. Four options are given in each Multiple-Choice Question based on List-I and List-II.
- Each set has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks:** +3 If **ONLY** the option corresponding to the correct combination is chosen
 - Zero Marks:** 0 If none of the options is chosen (i.e., the question is unanswered)
 - Negative Marks:** -1 In all other cases.

SECTION 2

- This section has **SIX (06)** Questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)**
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks:** +3 If **ONLY** the correct numerical value is entered;
 - Zero Marks:** 0 In all other cases.

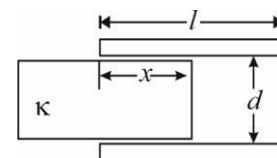
PART I : PHYSICS**62 MARKS****SECTION 1 | Type A**

This section consists of Eight (08) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

1. A circular beam of charge of radius a consists of electrons moving with a constant speed u along the $+z$ direction. The beam's axis is coincident with the z -axis and the electron charge density (in Coulomb/ m^3) is given by $\rho = cr^2$ where c is a constant and r is the radial distance from the axis of the beam. Which of the following statements is(are) true?

- (A) The current crossing the xy -plane is $\pi c u a^4$
 (B) The electric field at distance $r (r < a)$ from z axis is proportional to r^3
 (C) The magnetic field at distance $r (r < a)$ from z axis is proportional to r^3
 (D) The electric field at distance $r (r < a)$ from z axis is inversely proportional to r

2. Two square plates of sides l are placed parallel to each other with separation d as suggested in figure. You may assume d is much less than l . The plates carry uniformly distributed static charges $+Q_0$ and $-Q_0$. A block of metal has width l , length l , and thickness slightly less than d . It is inserted at distance x into the space between the plates. The charges on the plates remain uniformly distributed as the block slides in. In a static situation, a metal prevents an electric field from penetrating inside it. The metal can be thought of as a perfect dielectric, with $k \rightarrow \infty$.



- (A) The energy density of electric field between the charged plates in the region not occupied by the metal block is $\frac{Q_0^2}{(2l^4\epsilon_0)}$
 (B) The energy stored between plates as a function of x is given by $\frac{Q_0^2 d}{2l^2\epsilon_0}$
 (C) The energy stored between plates as a function of x is given by $\frac{Q_0^2 d(l-x)}{(2l^3\epsilon_0)}$
 (D) The magnitude of the force that acts on the metallic block is given by $\frac{Q_0^2 d}{2(l^3\epsilon_0)}$

3. A small object is placed on the axis of a convex lens of focal length f at a distance x from the lens. Let the distance between this object and the real image formed be d . Which of the following statements is(are) true?

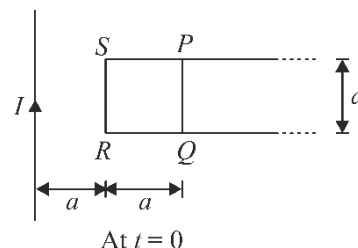
- (A) $d = \frac{x^2}{x-f}$
 (B) If $x > 3f$, then $d > 4.5f$
 (C) The minimum possible value of d is $4f$
 (D) If x is increased from $1.5f$ to $2f$, then d decreases from $4.5f$ to $4f$

4. An electron and an α particle, starting from rest, are accelerated through the same potential difference V . Let the kinetic energies gained by the particles be K_1 and K_2 , and let the velocities gained by them be v_1 and v_2 . Which of the following statements is(are) true?

[Assume that a proton and a neutron have equal mass, $\frac{\text{Mass of proton}}{\text{Mass of electron}} = 1836$]

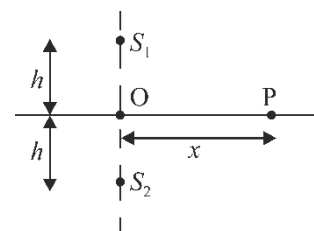
- (A) $\frac{K_1}{K_2} = \frac{1}{2}$ (B) $\frac{K_1}{K_2} = 1$ (C) $\frac{v_1}{v_2} = 42.9$ (D) $\frac{v_1}{v_2} = 60.6$

5. A long straight wire and a rectangular wire loop $PQRS$ are placed in the same plane with PQ and RS parallel to the straight wire. The straight wire carries a steady current I in the direction shown in the figure. The side PQ is a movable rod which can slide smoothly over the long rails. The rails have negligible resistance, and the side RS has resistance R_0 . Initially, $PQRS$ is a square of side length a and the distance between RS and the straight wire is kept fixed at a . Starting at $t = 0$, the rod PQ is moved towards the right at a constant velocity v . Which of the following statements is(are) true?



- (A) The magnetic flux through the loop $PQRS$ at $t = \frac{a}{v}$ is $\left(\frac{\mu_0 I a}{2\pi} \right) \log_e(3)$
- (B) The induced current in the loop $PQRS$ is anticlockwise in direction
- (C) The magnitude of the induced current keeps increasing with time
- (D) The magnitude of the induced current at $t = \frac{a}{v}$ is $\frac{\mu_0 I v}{6\pi R_0}$
6. Two coherent monochromatic point sources emitting light of wavelength λ (in vacuum) are placed on the y -axis and the origin is taken on the surface of water. One source is height h above the surface of water and the other source is depth h below the surface of water. Let x be the distance of a point P on the surface of water from the origin. If intensity due to each source at point P is I_0 , which of the following statements is(are) true? (n is a positive integer, refractive index of water $\mu = \frac{4}{3}$)

- (A) For $x = \sqrt{9n^2\lambda^2 - h^2}$, the intensity at point P is $4I_0$
- (B) For $x = [3n\lambda - h]$, the intensity at point P is $4I_0$
- (C) For $x = \left[\frac{3}{2}(2n-1)\lambda - h \right]$, the intensity at point P is zero
- (D) For $x = \frac{\sqrt{9(2n-1)^2\lambda^2 - 4h^2}}{2}$, the intensity at point P is zero



7. A sphere of radius R is uniformly charged over its volume with positive charge density ρ . The potential at a point on its surface is V_0 , and the potential at a point P , at a distance r from its centre, is $\frac{11}{8}V_0$. Which of the following options is(are) true?

(A) $r = \frac{2R}{3}$

(B) The magnitude of electric field at P is $\frac{V_0}{2R}$

(C) $\rho = \frac{3\epsilon_0 V_0}{4R^2}$

(D) Potential at all points inside the sphere is greater than V_0

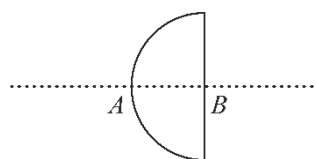
8. A solid hemisphere of radius R made out of glass of refractive index μ is fixed in air as shown. Light comes from a distant object parallel to AB , is incident on the curved face of the hemisphere, and is focused at a point P , at a distance d from B . Consider only paraxial rays. Which of the following options is(are) true?

(A) $d = \frac{R}{\mu - 1}$

(B) $d = \frac{R}{\mu(\mu - 1)}$

(C) The point P lies inside the hemisphere

(D) If the object is placed on the other side such that light coming parallel to AB now falls on the flat surface, it is focused at a distance $\frac{\mu R}{\mu - 1}$ from B



SPACE FOR ROUGH WORK

SECTION 1 | Type B

This section has TWO (02) List-Match Sets. Each List Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set – I for Question Numbers 9 & 10

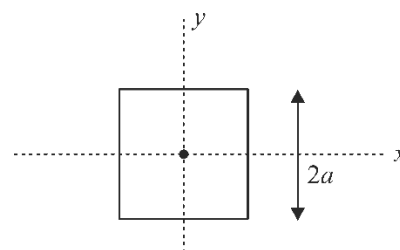
A rigid square loop of side length $2a$ is kept fixed in the x - y plane with its centre at the origin and a steady current I is made to flow in

the loop. A non-uniform magnetic field $\vec{B} = \left[B_0 \left(\frac{x}{a} \right)^m \right] \hat{k}$ exists,

where B_0 is a positive constant, and m is a positive integer. Let the

net force on the loop due to the magnetic field be $\vec{F}$.

Match List-I with List-II.



LIST-I		LIST-II	
P.	m is odd and the current I is clockwise	1.	$\vec{F} = 0$
Q.	m is even and the current I is clockwise	2.	$\vec{F}$ is along $+y$ direction
R.	m is odd and the current I is anti-clockwise	3.	$\vec{F}$ is along $-y$ direction
S.	m is even and the current I is anti-clockwise	4.	$\vec{F}$ is along $+x$ direction
		5.	$\vec{F}$ is along $-x$ direction

9. Which of the following is correct?

(A) P - 1 (B) Q - 1 (C) R - 5 (D) S - 4

10. Which of the following is correct?

(A) P - 5 (B) Q - 3 (C) R - 2 (D) S - 5

SPACE FOR ROUGH WORK

List Match Set – II for Question Numbers 11 & 12

Match the following List – I with List II. Take Planck's constant $h = 6.6 \times 10^{-34}$ J-s, $hc = 1240$ eV-nm, mass of electron $m_e = 9 \times 10^{-31}$ kg, charge of electron $e = 1.6 \times 10^{-19}$ C.

LIST-I			LIST-II
P.	Cut-off wavelength of continuous X-rays from a discharge tube with an applied potential difference 20 kV	1.	1.32 nm
Q.	de Broglie wavelength of the maximum energy photoelectron ejected from a metal surface of work function $\phi_0 = 7.4$ eV when light of wavelength 100 nm is incident on the metal surface	2.	0.062 nm
R.	Wavelength of the second lowest energy line in the Balmer series of the Hydrogen spectrum	3.	486.3 nm
S.	de-Broglie wavelength of an electron in the 4 th Bohr orbit in the Hydrogen atom	4.	0.55 nm

- 11.** Which of the following is a correct combination?
(A) P – 1, Q - 4 **(B)** P – 2, Q - 1 **(C)** P – 2, Q - 4 **(D)** P – 2, Q - 3
- 12.** Which of the following is a correct combination?
(A) R – 3, S - 2 **(B)** R – 3, S - 1 **(C)** R – 2, S - 1 **(D)** R – 3, S - 4

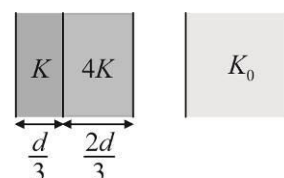
SPACE FOR ROUGH WORK

SECTION 2

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

1. Consider a planar rectangular surface in the x - y plane with its vertices at $A(0, a, 0)$, $B(a, a, 0)$, $C(a, -a, 0)$ and $D(0, -a, 0)$. Two point charges, $+Q$ and $-Q$ are kept at $(0, 0, a)$ and $(0, 0, -a)$. If the net electric flux through the rectangular surface is $n \frac{Q}{\epsilon_0}$, then the value of n is _____.

2. Two identical capacitors of plate area A and plate separation d initially have vacuum between the plates. Now, two dielectric slabs of dielectric constant K and $4K$ and widths $\frac{d}{3}$ and $\frac{2d}{3}$ are inserted in the first capacitor, and a single dielectric slab of dielectric constant K_0 and width d is inserted into the second capacitor. Area of the dielectric slabs is equal to the plate area A .



After the insertion, the capacitance of the two capacitors is equal. Then, $\frac{K_0}{K} =$ _____.

3. A series circuit consists of a resistance $R = 100 \Omega$, a coil of inductance $L = 200 \text{ mH}$, a capacitor of capacitance $C = 100 \mu\text{F}$ and an AC source of angular frequency $\omega = 200 \text{ rad/s}$. If the rms value of the potential difference across the capacitor is 40 V , the rms value of the potential difference across the resistance is _____ V.

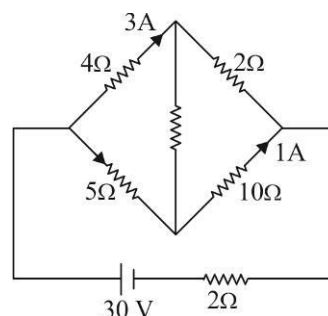
4. A radioactive isotope of iodine, I-131 , decays into a stable isotope of Xenon, Xe-131 , in two steps, emitting a beta particle and gamma radiation as follows:



The maximum energy of the beta particle emitted is 606.623 keV . The wavelength of the highest energy gamma photon that can be emitted is _____ $\times 10^{-12} \text{ m}$. [Mass of ${}^{131}_{53}\text{I}$ atom = 130.906124 amu , mass of ${}^{131}_{54}\text{Xe}$ atom = 130.905082 amu , $1 \text{ amu} = 931.5 \text{ MeV} / c^2$, $hc = 1242 \text{ eV} \cdot \text{nm}$]

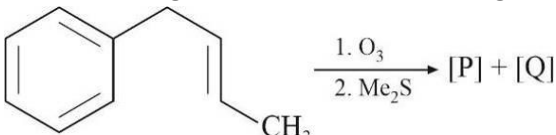
5. A parallel beam of intensity 3000 W/m^2 , $\lambda = 100 \text{ nm}$ having cross sectional area 100 cm^2 is incident normally on a glass slab, placed in air. The slab reflects 40% of light, transmits 40% of light and absorbs 20% of light incident on it. The force experienced by slab due to incident radiation beam will be $x \times 10^{-y} \text{ N}$. Find the value of $x + y$. _____

6. In the given circuit, currents through 4Ω and 10Ω resistances are as shown. The current through the 5Ω resistance is _____ A.



PART II : CHEMISTRY**62 MARKS****SECTION 1 | Type A**

This section consists of Eight (08) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- Which of the following is/are main forces which stabilise the secondary and tertiary structures of proteins ?
 (A) Hydrogen bonds (B) Disulphide linkages
 (C) Vander Waal's forces (D) Electrostatic forces of attraction
- Which of the following is/are correct ?
 (A) violet colouration of iodine is due to HOMO to LUMO charge transfer
 (B) violet colouration of KMnO_4 is due to ligand to metal charge transfer
 (C) violet colouration of $[\text{Ni}(\text{en})_3]^{2+}(\text{aq})$ is due to d-d transition
 (D) violet colouration of solid ozone is due to absorption of visible radiations
- Which of the following is/are correct for reaction given below ?

 (A) It is reductive ozonolysis reaction
 (B) one of the organic product of this reaction undergo haloform reaction
 (C) (P) and (Q) on heating in presence of NaOH produce 2-phenyl but-2-ene
 (D) configuration of $\text{C}=\text{C}$ of side chain is Z.
- During cation analysis, group-I metal ions are precipitated as chlorides. Which of the following statements is/are true regarding the group-I radicals analysis ?
 (A) Ag^+ , Hg_2^{2+} and Pb^{2+} form insoluble chloride with the addition of dil. HCl
 (B) Pb^{2+} does not precipitate in hot conc. HCl
 (C) AgCl dissolve in aqueous NH_3 and will not reprecipitate with the addition of dil. HCl
 (D) Hg_2Cl_2 undergoes disproportionation on reaction with NH_3 .
- Which of the following is/are correct regarding $\text{K}_2\text{Cr}_2\text{O}_7$?
 (A) In a redox titration using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ as oxidising agent, diphenyl amine is used as indicator
 (B) When heated with NaCl in presence of conc. H_2SO_4 it produce red vapours of chromyl chloride
 (C) Orange colouration of aq. $\text{K}_2\text{Cr}_2\text{O}_7$ is due to d-d transition
 (D) Its solution becomes yellow coloured when made alkaline

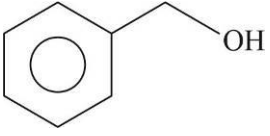
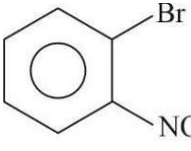
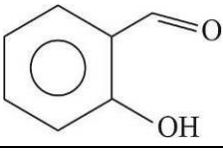
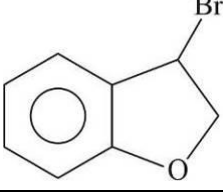
6. Which of these is/are correct statement(s) about the ferrioxalate ion $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$?
(A) It is chiral
(B) There are two different carbon-oxygen bond distances in the ion
(C) There are six iron-oxygen bonds in the ion
(D) Five moles of ferrioxalate ions are oxidised by six moles of acidified MnO_4^- ion
7. Which of the following statement(s) is/are correct?
(A) Disulphur difluoride exists as two structural isomers
(B) Sulphur difluoride, SF_2 , is very unstable, dimerize to disulphur tetrafluoride, S_2F_4 , in which all four fluorines are in different environments
(C) Sulphur tetrafluoride is less volatile than sulphur hexafluoride
(D) Sulphur hexafluoride is almost chemically inert
8. Which of the following unit cell has $\alpha = \beta = \gamma = 90^\circ$?
(A) Cubic (B) Tetragonal (C) Orthorhombic (D) Rhombohedral
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SPACE FOR ROUGH WORK

SECTION 1 | Type B

This section has TWO (02) List-Match sets. Each List Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.

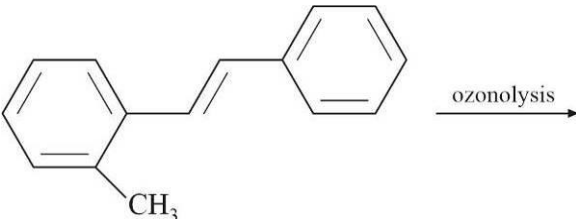
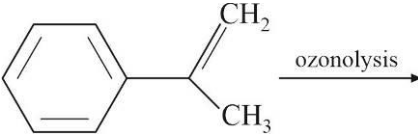
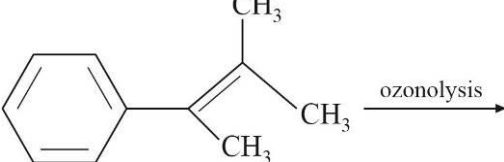
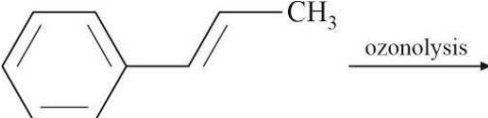
List Match Set – I for Question Numbers 9 & 10

List-I (Organic Compound)		List-II (Reaction Characteristics)	
(P)		I.	Dehydrogenation
(Q)		II.	Nucleophilic substitution reaction
(R)		III.	Nucleophilic addition reaction
(S)		IV.	Dehydrohalogenation
		V.	Oxidation by Tollen's reagent
		VI.	Reduction by LiAlH_4

9. Which of the following combination of reactant and reaction shown is correct ?
 (A) P - II, V (B) Q - II, VI (C) R - I, III (D) S - III, VI
10. Which of the following combination of reactant and reaction shown is not correct ?
 (A) P - I, II (B) Q - II, IV (C) R - III, V (D) S - I, IV

SPACE FOR ROUGH WORK

List Match Set – II for Question Numbers 11 & 12

List-I (Reductive Ozonolysis)		List-II (Characteristics of Products)	
(P)		I.	All product undergoes cannizzaro reaction
(Q)		II.	All product undergoes haloform reaction
(R)		III.	All product form silver mirror with Tollen's reagent
(S)		IV.	All product shows aldol reaction
		V.	All product do not reacts with Fehling's solution

11. Which of the following is correct ?

- (A) P - I, III (B) Q - II, IV (C) R - I, IV (D) S - I, III

12. Which of the following is correct ?

- (A) P - IV (B) Q - I (C) R - II (D) S - V

SPACE FOR ROUGH WORK

SECTION 2

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25)

1. 0.5gm benzaldehyde and ethanal mixture when heated with Fehling solution gave 1.43gm of red precipitate. What is percentage of ethanal in the mixture? [Atomic mass Cu = 63.5, O = 16, C = 12, H = 1]
2. 0.90g of an organic compound $C_4H_{10}O_2$ when treated with Na gives 224mL of H_2 at STP. How many isomers including stereoisomers are possible for $C_4H_{10}O_2$? [Atomic mass O=16, C=12, H=1]
3. A purified pepsin isolated from a bovine preparation was subjected to an amino acid analysis. The amino acid present in smallest amount was lysine, $C_6H_{14}N_2O_2$ and the amount of lysine was found to be 0.73g per 100g protein. What is the minimum molecular weight of protein? [Atomic mass O = 16, N = 14, C = 12, H = 1]
4. The gas phase decomposition of P at 1500K follows two decomposition paths.

Elementary process	Rate constant
(i) $P \longrightarrow Q + R$	$3s^{-1}$
(ii) $P \longrightarrow M + N$	$5s^{-1}$

What is the maximum theoretical percentage yields of M at 1500 K ?
5. What is the molar mass of a compound that has a concentration $\omega = 1.2g/L$ in solution and an osmotic pressure, $\pi = 0.20atm$ at $T = 300K$? [$R = 0.0821 L atm mol^{-1} K^{-1}$]
6. What is atomic packing factor (packing fraction) of the diamond cubic structure? [$\sqrt{3} = 1.732$, $\pi = 3.14$]

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**62 MARKS****SECTION 1 | Type A**

This section consists of Eight (08) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- If $f(x) = \lim_{n \rightarrow \infty} n \left(x^{\frac{2017}{n}} - x^{\frac{2018}{n}} \right)$; $x > 0$ then choose the correct option(s) :

(A) the number of solution(s) of the equation $f(|x|) = e^{-2018x^2}$ is 2
 (B) the number of solution(s) of the equation $f(|x|) = e^{2018x^2}$ is 2
 (C) the number of solution(s) of the equation $f(x) = e^{2017}$ is 1
 (D) the number of solution(s) of the equation $f(x) = e^{-2017}$ is 1
- Consider a 3×3 matrix A whose element $a_{ij} = \tan^{-1}(\tan(i+j))$, $1 \leq i, j \leq 3$. Which of the following statement(s) is/are correct ?

(A) A is a symmetric matrix (B) $\text{trace}(A) = 12 - 4\pi$
 (C) $\text{trace}(A) = 10 - 4\pi$ (D) $\text{Adj}(A) + (\text{Adj}(A))^2$ is a symmetric matrix
 (where $\text{Adj}(A)$ denotes adjoint of a matrix A and $\text{trace}(A)$ denotes sum of diagonal entries of a matrix)
- If $f(x) = \sqrt{1-x^2}(\cos^2(\tan^{-1}x) - x^2 \sin^2(\tan^{-1}x))$, then :

(A) Range of $f(x)$ is R (B) Range of $f(x)$ is $[0, 1]$
 (C) $\int_0^1 f(x)dx$ is $\frac{3\pi}{16}$ (D) $\int_0^1 f(x)dx$ is $\frac{3}{8}$
- Let $f : R \rightarrow (0,1)$ be a continuous function, then which of the following function(s) has/have the value zero at some point in the interval $(0, 1)$?

(A) $f(x) + \int_0^{\pi/2} f(t) \tan t dt$ (B) $x^{11} - f(x)$
 (C) $e^x - \int_0^x f^2(t)dt$ (D) $e^x + \int_0^x f(t)dt$
- If $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} \tan(\sin(\tan^{-1}x)) + 2\sqrt{1-x^2} \sin(\tan(\sin^{-1}x)) - 3x}{x^p}$ exists and is non-zero, then choose the correct option(s).

(A) The value of p is 3 (B) The value of p is 5
 (C) The value of limit is $\frac{-31}{60}$ (D) The value of limit is $\frac{-8}{15}$
- A is 3×3 matrix whose every entry is 1 or -1 . Choose the correct option(s).

(A) The maximum value of determinant of A is 4.
 (B) Total number of different values of determinant of A can take is 3.
 (C) Total number of different values of determinant of A can take is 5.
 (D) Total number of different matrices A , such that determinant of A is positive, is 96.

7. If $L = \lim_{n \rightarrow \infty} \frac{n^3 (e^{1/n} + e^{2/n} + \dots + e)}{(n+1)^m (1^m + 4^m + \dots + n^{2m})}$ is non-zero finite real, then:
- (A) $L = 3(e-1)$ (B) $L = e-1$ (C) $m=1$ (D) $m = \frac{4}{3}$
8. $f(x)$ is a four-degree polynomial with leading coefficient unity and $f(1)=1$, $f(2)=4$, $f(3)=9$. If $x=1$ is a point of extremum of $f(x)$, then:
- (A) $f(4)=16$ (B) $f'(2)=4$
(C) $f''(x)=2$ has atleast one solution in $(0, 4)$ (D) $f'(2)=0$
-

SPACE FOR ROUGH WORK

SECTION 1 | Type B

This section has TWO (02) List-Match sets. Each List Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set – I for Question Numbers 9 & 10

Match the following List-I with List-II

List-I		List-II	
P.	$ A_{2 \times 1} B_{1 \times 2} $ is equal to	1.	Data insufficient
Q.	If A and B are non-zero square matrices of same order such that $AB=0$, then the value of $ A + B +5$ is equal to	2.	$2/27$
R.	Let A be an 3×3 orthogonal matrix such that $AB=BA$, then $ AB^T - B^T A + 2I $ is equal to	3.	0
S.	Let A and B be square matrices of order 3×3 such that $ A =1$ and $ B =3$, then the value of $ 2A^2 B^{-3} $ is equal to	4.	8
		5.	5
		6.	$8/27$

9. Which of the following is correct combination?
 (A) $P \rightarrow 1, Q \rightarrow 5$ (B) $P \rightarrow 1, Q \rightarrow 1$ (C) $P \rightarrow 3, Q \rightarrow 5$ (D) $P \rightarrow 3, Q \rightarrow 1$
10. Which of the following is correct combination?
 (A) $R \rightarrow 4, S \rightarrow 6$ (B) $R \rightarrow 4, S \rightarrow 2$ (C) $R \rightarrow 1, S \rightarrow 2$ (D) $R \rightarrow 1, S \rightarrow 6$

List Match Set – II for Question Numbers 11 & 12

Let A and B are points on the line L_1, L_2 respectively such that $d = OA + OB + AB$ is minimum (O being the origin), match the following List-I with List-II.

List-I		List-II	
P.	$L_1: x-1=y-1=\frac{z+1}{0}, L_2: x+2=4-y=\frac{z+1}{0}$	1.	$d=4\sqrt{2}$
Q.	$L_1: \frac{x-1}{2}=1-y=1-z, L_2: x-2=y-2=\frac{2-z}{2}$	2.	$d=2\sqrt{3}$
R.	$L_1: x=y=z-\sqrt{2}, L_2: x=y=z+\sqrt{2}$	3.	$d=6\sqrt{2}$
S.	$L_1: 2x-3=6-y=\frac{z}{0}, L_2: x+3=2y=\frac{z}{0}$	4.	$d=4\sqrt{3}$

11. Which of the following is correct combination?
 (A) $P \rightarrow 2, Q \rightarrow 4$ (B) $P \rightarrow 2, Q \rightarrow 2$ (C) $P \rightarrow 4, Q \rightarrow 4$ (D) $P \rightarrow 4, Q \rightarrow 2$
12. Which of the following is correct combination?
 (A) $R \rightarrow 1, S \rightarrow 1$ (B) $R \rightarrow 1, S \rightarrow 3$ (C) $R \rightarrow 3, S \rightarrow 1$ (D) $R \rightarrow 3, S \rightarrow 3$

SECTION 2

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

- Given the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ and satisfies $A^3 = pA^2 + qA + rI$ where (p, q, r) are constants & (I is a unit matrix of order 3×3), then the value of $(p + q + r)$ is :
- Let $f(x) = \lim_{n \rightarrow \infty} \frac{3^n \sin x + (\sqrt{2} \cos x + 2)^n + 2^n \cos x}{3^n + \cos x (\sqrt{2} \cos x + 2)^n}$, then the value of $\left(\lim_{x \rightarrow \frac{\pi^-}{4}} f(x) \right)^4$ is equal to ____
- $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function satisfying $2(f(x))^2 f'(x) = (f(2x+1))^2 f'(2x+1)$, $f\left(-\frac{1}{8}\right) = 1$, $f(6) = 2$ find the value of $\frac{2}{(f(-1))^3}$.
- A point $P\left(\frac{e^t + e^{-t}}{2}, \frac{e^t - e^{-t}}{2}\right)$ traces a locus $S = 0$ in X - Y plane (t being a non-negative parameter). A fixed point P_0 , having parameter t_0 , lies on the curve $S = 0$. The area bounded by the curve, line OP_0 and x -axis is 240 sq. units (O being the origin), find the value of t_0 .
- Given that $\begin{vmatrix} a & d & 1 \\ b & e & 1 \\ c & f & 1 \end{vmatrix} = -5$, $\begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ d & e & f \end{vmatrix} = -3$ find the value of $\begin{vmatrix} c & 3 & f \\ b & 1 & e \\ a & -1 & d \end{vmatrix}$.
- The curve $f(x, y) = 0$ passing through $(0, 2)$ satisfy the differential equation $\frac{dy}{dx} = \frac{y^3}{e^x + y^2}$. If the line $x = \ln 5$ intersect it at points $y = \alpha$ and $y = \beta$, then find the value of $\alpha^2 + \beta^2 + \alpha\beta$.

SPACE FOR ROUGH WORK